Parameter-Efficient Fine-Tuning

for Czech-to-English Machine Translation

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Motivation & Research Question

01 **Problem:** Full fine-tuning is computationally expensive, hard to scale across language pairs and risks overfitting.

02 **Setting:** We evaluate Czech -> English translation with NLLB-200-distilled-600M in a low-resource setting. Czech is challenging because of rich morphology and flexible word order.

03 **Approach:** PEFT methods are lightweight. LoRA, IA3, and Prefix tuning keep the base model frozen while training less than 0.2% of the parameters.

04 **Goal:** Evaluate whether PEFT methods preserve translation quality while substantially reducing the number of trainable parameters.

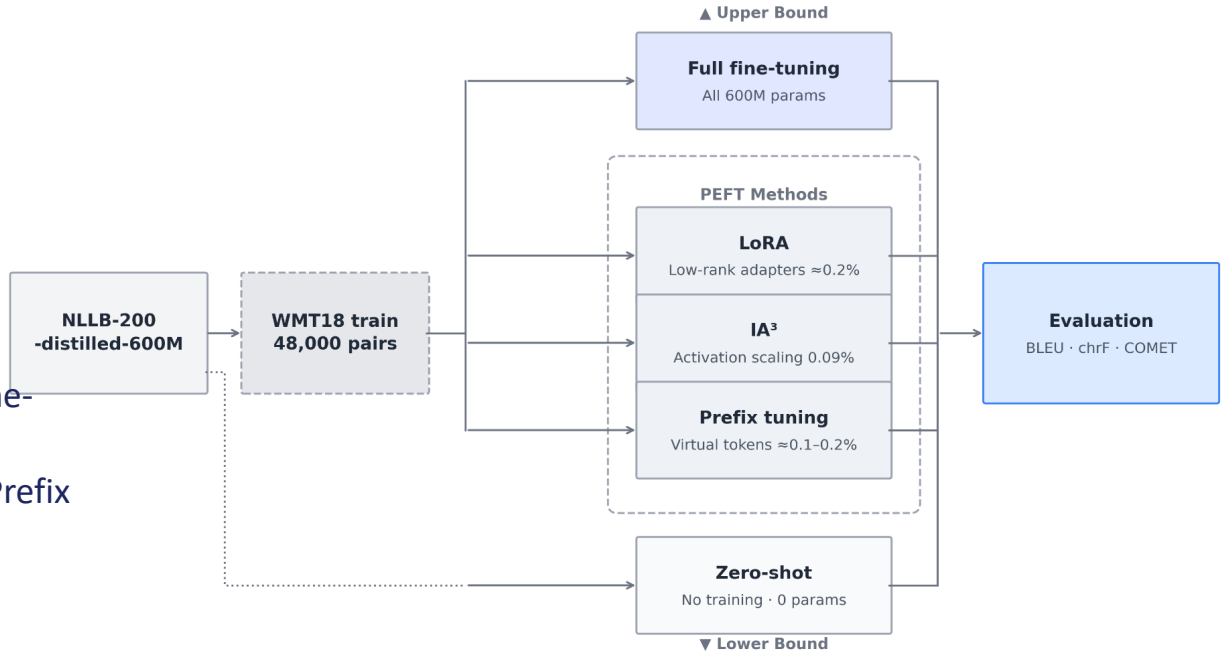
RESEARCH QUESTION

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How do LoRA, IA3, and prefix tuning compare to full fine-tuning and zero-shot for Czech→English MT in a low-resource setting?

Experiment Framework

- **Same base model**
- **Same dataset**
  - WMT18 Czech-English
  - 48k/1k/1k split
- **Same evaluation metrics**
  - BLEU, chrF, COMET
- **Five pipelines**
  - Baselines: Zero shot, full fine-tuning
  - PEFT methods: LoRA, IA3, Prefix tuning



Results

System	Trainable params	BLEU	COMET	chrF
Zero-shot	0	23.04	0.741	50.11
LoRA	1.18M(0.19%)	**31.70** best BLEU	0.831	**57.82**
IA^3	560k (0.09%)	30.26	0.828	56.83
Prefix tuning	491K (0.08%)	30.55	0.828	56.98
Full fine-tuning	600M (100%)	31.15	**0.833** best COMET	57.47

PEFT methods match full fine-tuning with **<0.2% of parameters**
slightly exceeded in some cases

LoRA Fine-tuning

METHOD

Pretrained weights stay frozen. Two small decomposed matrices A and B are injected in selected attention layers of the transformer's architecture. Only these are updated during training, then they are merged at inference time.

```
ΔW = B × A              rank r controls
capacity

h = Wx + BAx
```

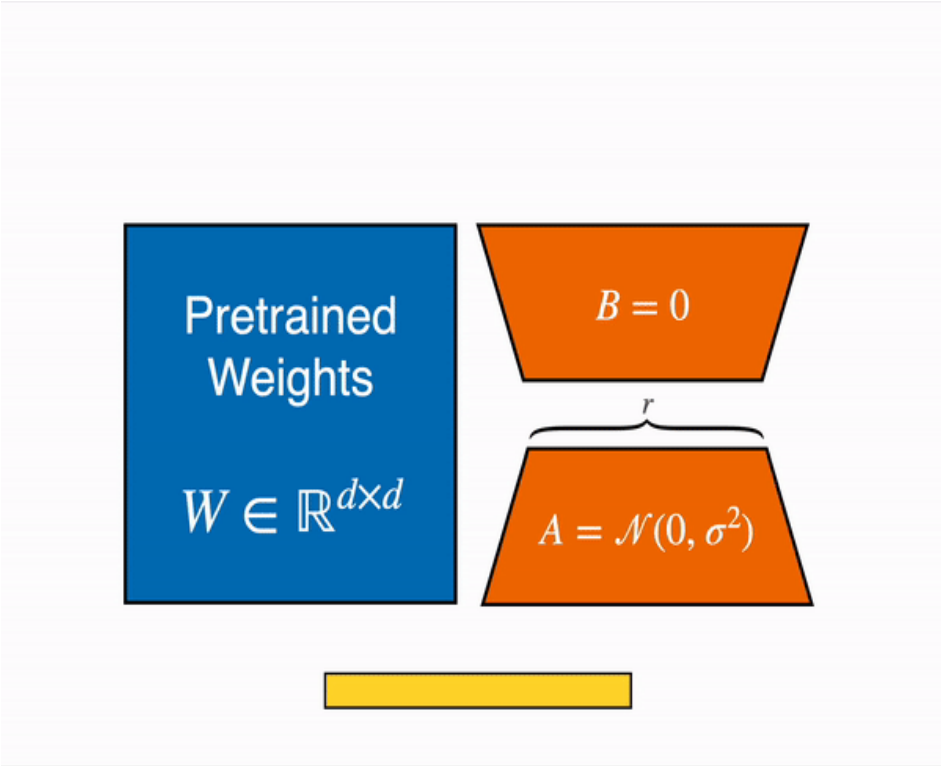


Fig 1: Medium.com, *Understanding Low-Rank Adaptation (LoRA)*

LoRA Fine-tuning

TAKEAWAYS

Trainable parameters (rank=8)

1.18M 0.19% of model

BLEU	chrF	COMET
31.70	**57.82**	**0.831**

Learning rate

`LR` 3e-4 was best after 1 epoch, but worse after 3 epochs.

LoRA has fewer trainable parameters, so optimization behaves differently from full fine-tuning.

LoRA Rank vs. alpha

LoRA Rank controls the **capacity** of the LoRA adapter by setting the **size** of the low-rank matrices.

Alpha hyperparam. determines the **scaling factor** (the multiplier applied to weight changes): *`alpha/r`*

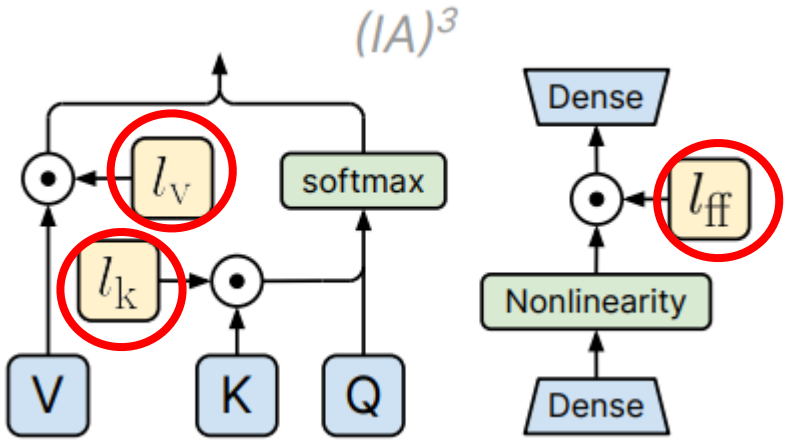
BLEU vs COMET

The best BLEU checkpoint was not the best COMET checkpoint, suggesting **lexical** quality and **semantic** quality peaked at different points.

IA^3 （Infused Adapter by Inhibiting and Amplifying Inner Activations） Fine-tuning

What is IA^3?

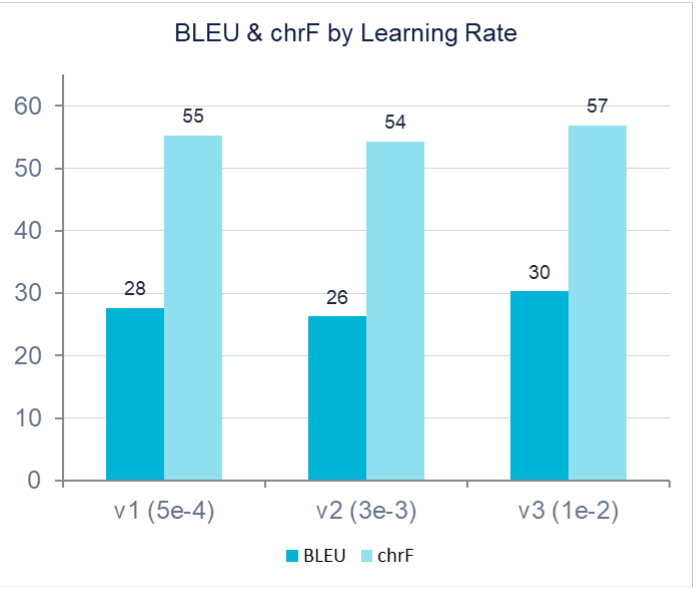
- Add **learnable scaling vectors**
- Modulate **attention** and **feed-forward activations** via element-wise scaling
- Fine-tune only a **small number of parameters** for efficient adaptation 560k (0.09%)



Liu et al. (2022)

IA³ (Infused Adapter by Inhibiting and Amplifying Inner Activations) Fine-tuning

Experimental Results



COMET Scores

v1 · LR = 5e-4
0.8118 2nd
BLEU 27.61

v2 · LR = 3e-3
0.8070 3rd
BLEU 26.24

v3 · LR = 1e-2
0.8280 Best
BLEU 30.26

TAKEAWAYS

- **Higher LR benefits IA³**

IA³ tolerates LR as high as 1e-2 without overfitting (Liu et al., 2022).

- **Non-monotonic LR response**

PEFT hyperparameter landscapes are non-monotonic with sharp drops at suboptimal values

- **High train loss ≠ worse eval**

V3 has highest train loss (44.67) yet best BLEU/COMET.

Prefix Tuning

Method principle: virtual tokens, frozen model

Trainable parameters: 491k (0.08%) vs. 1.2M (0.20%)

Key finding: The simplest configuration was the best trade-off

Error Analysis | Linguistic-specific

Feature	Source (cs)	Reference (en)	System output	Observation
Proper nouns & named entities	Na **Kolínsku** hořelo pole.	A field was burning in **Cologne**.	Fire breaks out in Kolín.	**Prefix & IA3** – Kolínsko (Czech region) mistaken for Cologne, Germany
Ellipsis	Ten s dvaceti chudý.	The poor guy with 20.	Those earning 20,000 are poor.	**All systems** – verb and currency unit missing from source
Idiomatic expressions	Olej do ohně přilévá chudoba.	Olej do ohně přilévá chudoba.	Poverty feeds the discontent.	**Full FT** – idiom copied verbatim, not translated
False friends	**Fandí** všichni.	They're all **fanatics**.	Everyone supports them.	**IA3** – fandí (to cheer for) confused with fanatik (fanatic)
Missing discourse context	Jeho podmořská **hrobka** zachovala tělo v dokonalém stavu.	His underwater **tomb** kept the body in perfect condition.	But the dinosaur's undersea burial preserved its armor in exquisite detail.	**All systems** - sentence about a dinosaur fossil, not a human

Other examples



Lost in translation - literal nonsense

SRC : Volič tápe i šířeji. (Full FT)
HYP : **Tapestry volley and spread.**
REF : The voter is more in the dark than ever.

Tápe means "fumbles/gropes in the dark" → the model apparently saw "tap" and went with tapestry.

The confident wrong answer



SRC : Dám vám **hádanku**. (Full FT)
HYP : I'll give you my **back**.
REF : Here's a puzzle for you.

Hádanku = *riddle*. The model decided it means "my back". Very confident, very wrong.

Other examples



The existential crisis

SRC : A moje **úvodní řeč**? (Full FT)
HYP : What's my **first language**?
REF : As for my captain's speech?

A question about an **opening speech** turned into an identity crisis



The supporter becomes a fanatic

SRC : **Fandí** všichni. (IA3)
HYP : They're all **fanatics**.
REF : Everyone supports them.

Fandí = to cheer for/support.
IA3 turned a wholesome crowd of supporters into a mob of fanatics.

Conclusion

- **PEFT methods match or closely approach full fine-tuning quality** for Czech-to-English translation, achieving competitive BLEU, chrF, and COMET scores while updating only a tiny fraction of model parameters.

- **Parameter efficiency:** LoRA, IA^3, and Prefix Tuning update **less than 0.2%** of the parameters in NLLB-200-distilled-600M, substantially reducing training costs and memory requirements compared to full fine-tuning.

- **Practical takeaway:** For low-resource adaptation and limited computational budgets, **PEFT methods provide a highly effective alternative to full fine-tuning**. Among the methods tested, **LoRA offers the strongest overall performance**, while **Prefix Tuning provides an attractive balance between translation quality and parameter efficiency**.

Questions?